

■ EcoSynapse™ Intelligent Platform

Complete Technical Documentation, System Architecture, Algorithms & Specifications

1. Executive Summary & System Overview

EcoSynapse is an end-to-end IoT, Artificial Intelligence, and cloud-powered smart waste management ecosystem designed for modern residential communities, commercial hubs, and smart cities. The platform automates waste segregation verification, incentivizes residents through an algorithmic **EcoPoints** rewards system, optimizes sanitation collector dispatch routes using predictive telemetry, and delivers real-time fleet analytics to facility managers.

Table 1.1: Role-Based Access Control (RBAC) Matrix

Stakeholder Role	Access Scope	Key System Capabilities
Resident / Owner	activeRole === 'user'	• View EcoPoints balance, tier status & rank • Scan Bin QR code & simulate verified dry disposals • Redeem EcoPoints for utility/shopping rewards • View assigned bin status notifications (Read-Only)
Sanitation Collector	activeRole === 'collector'	• View priority collection route sorted by fill urgency (≥80%) • Receive live audio voice notifications for critical bins • Filter urgent bins vs all community nodes • Mark bins as emptied & log route completion
Facility Admin Manager	activeRole === 'admin'	• Monitor community-wide waste telemetry & carbon metrics • ESP32 Hardware Simulator (Inject overfill/contamination) • ESP32-CAM AI Snapshot inspection feed • System audit log monitoring & node connection toggles

2. 3-Tier System Architecture

The EcoSynapse architecture is partitioned into three decoupled functional layers:

- **Layer 1: Smart Bin Edge Layer:** ESP32 microcontroller controller, ESP32-CAM optical module for AI computer vision classification, dual ultrasonic fill level sensors (Dry Recyclables and Wet Organic), strain-gauge load cell (waste weight in kg), capacitive moisture grid (contamination detection), bottom liquid sump sensor, and ToF (Time-of-Flight) proximity sensor for touchless lid opening.
- **Layer 2: Core Cloud Layer:** Next.js 14 API Gateway (/api/bins, /api/chat), Supabase PostgreSQL relational storage, WebSockets real-time sync channel, EcoScore Engine, and Google Gemini AI Engine.

- **Layer 3: Application Layer:** Resident Portal (Scan & Earn), Collector App (Priority Dispatch & Route Urgency), and Admin Dashboard (Telemetry Simulator).

3. End-to-End Workflows

Workflow 3.1: Waste Disposal & Verified Rewards Loop

1. Resident approaches the Smart Bin Node; ToF lid proximity sensor detects hand and automatically opens lid.
2. User deposits waste into designated compartment.
3. ESP32-CAM module captures high-resolution intake image snapshot while load cell measures weight addition (kg).
4. Edge sensors check moisture content. If dry waste compartment moisture exceeds 65%, a `CONTAMINATION_ALERT` is flagged.
5. Telemetry packet transmitted to Next.js cloud server via HTTPS/MQTT.
6. Upon verified disposal, user wallet is credited with +10 EcoPoints, EcoScore is recomputed, and audio voice confirmation plays.

Workflow 3.2: Sanitation Collector Priority Dispatch & Routing

1. Telemetry background engine continuously evaluates net fill percentages across all smart bins.
2. When any bin reaches $\geq 80\%$ capacity or flags a sump leak, it is elevated to **PRIORITY DISPATCH REQUIRED** status.
3. Live WebSocket notification triggers a red notification badge in the header bell and plays an automated voice alert to sanitation drivers.
4. The Collector App orders stops automatically by fill urgency percentage.
5. Upon collection, the driver taps **Mark Emptied**, resetting bin capacity to 0% and logging the audit event.

4. Core Algorithms & Mathematical Formulas

4.1 EcoPoints Transaction Formula (EP)

Each waste disposal transaction generates EcoPoints (EP) calculated using the weighted multi-factor formula:

$$EP = (SA \times 40\%) + (RP \times 25\%) + (WR \times 20\%) + (CS \times 10\%) + (CB \times 5\%)$$

Formula Variable Definitions:

- **SA (Segregation Accuracy Score, 0-100):** Computer vision confidence and moisture sensor grid score.
- **RP (Recycling Participation Score, 0-100):** Net weight and deposit frequency of dry recyclable materials.
- **WR (Waste Reduction Trend Score, 0-100):** Evaluates historical reduction in non-recyclable waste volume.
- **CS (Consistency Score, 0-100):** Daily consecutive disposal streak bonus (+2 to +10 points).
- **CB (Correct Disposal Bonus, 0-100):** Special bonus for proper e-waste or hazardous waste segregation.

4.2 User Tiers & EcoScore Multipliers

Final awarded EcoPoints are scaled by the user's monthly EcoScore tier status:

Final EcoPoints = EP × Tier Multiplier			
User Tier	Monthly EcoScore Range	Point Multiplier	Tier Benefits
■ Green	< 50 Score	1.0x	Standard access to rewards catalog
■ Silver	50 – 69 Score	1.1x	10% higher points earning, early campaign access
■ Gold	70 – 89 Score	1.25x	25% higher points earning, exclusive lifestyle perks
■ Platinum	≥ 90 Score	1.5x	50% higher points earning, AGM green plaque award

4.3 Predictive Overfill Velocity Algorithm

The predictive telemetry engine estimates hours remaining before bin overflow:

$\Delta \text{ FillPerHr} = (\text{OverallFill_t2} - \text{OverallFill_t1}) / (\text{t2} - \text{t1})$
$\text{Predicted Hours} = \text{Max}(0.1, (100 - \text{CurrentOverallFill}) / \Delta \text{ FillPerHr})$

5. Database Schemas & Technology Stack

5.1 Data Model Summary

- **SmartBin:** Tracks id, name, location, dryFill, wetFill, overallFill, battery, weight (kg), moistureLevel, isOnline, hasContamination, hasLiquidLeak, predictedFullHours, lastUpdated.
- **UserProfile:** Tracks id, name, userType ('Resident'|'Owner'), flatNo, ecoPoints balance, ecoScore, totalDisposals, rank, totalResidentsInApartment.
- **RewardItem:** Vouchers catalog (Instant UPI Cash, Plant a Tree, Metro/Bus Pass, Electricity Discount, Apartment Maintenance Credit).
- **SystemEvent:** Audit trail for overfill warnings, sensor leaks, and verified disposal events.

5.2 Technology Stack Summary

- **Frontend & App Framework:** Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, Lucide React.
- **Backend Services:** Next.js Server Actions & API Routes (/api/bins, /api/chat).
- **Database & Realtime:** Supabase PostgreSQL + Auth Service with Hybrid Fallback Architecture.
- **Artificial Intelligence:** Google Gemini API Engine (@google/genai SDK) for EcoBot AI Assistant.
- **Audio Engine:** Native Web Speech Synthesis API (window.speechSynthesis) for automated voice alerts.